

Efficient Quantum Algorithm for Identifying Hidden Polynomial Function Graphs

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Joint work

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Motivation and Context

- Which problems can be solved exponentially faster on quantum computer than on a classical computer?
- Most such problems are black-box problems
 - Hidden Subgroup Problems (abelian/nonabelian)
 - Hidden Shift Problems
- Motivation: interesting problems can be reduced to these black-box problems
 - Integer Factorization, Discrete Logarithm
 - Pell's Equation, Principle Ideal Problem, Problem of Computing the Unit and Class Group of a Number Field
 - Shortest Lattice Problem
 - Graph Isomorphism

Motivation and Context

- Most quantum algorithms rely on the efficient solution of the abelian HSP
- The nonabelian HSP seems to be hard for groups of interest such as the symmetric and dihedral groups
- Are there other ways of extending the success of the abelian HSP besides considering the nonabelian case?

Non-linear Structures

Design quantum algorithms for certain black-box problems involving **non-linear structures**

A. Childs, L. Schulman, and U. Vazirani, *Quantum algorithms for hidden nonlinear structures*, QIP 2007

- Query complexity of Hidden Polynomial Problem
- Efficient quantum algorithms for Hidden Sphere Problems
 - Hidden Radius Problem
 - Hidden Subspace of Centers Problem

Outline

- Definition of the Hidden Polynomial Function Graph Problem
- Reduction to quantum state identification
- Polynomial quantum query complexity
- Construction of a von Neuman measurement for identifying the quantum states
 - Reduction of its implementation to a classical algebro-geometric problem
 - Efficient solution of this problem

Definition

- let $Q(X) \in \mathbb{F}[X]$ be an arbitrary polynomial of total degree at most n whose constant term is equal to 0
- let $B : \mathbb{F}^2 \rightarrow \mathbb{F}$ be a black-box function hiding the polynomial Q in the following sense:

$$B(x, y) = \pi(y - Q(x)) ,$$

where π is an unknown bijection permuting the elements of $\mathbb{F}$ arbitrarily

- the Hidden Polynomial Function Graph Problem is to identify Q
- an algorithm for polynomials with degree less or equal to n (where n is a constant) is efficient if its running time is polylogarithmic in p , i.e., the size of $\mathbb{F}$

Hidden Polynomial Function Graph

can be considered as a natural generalization of abelian HSP over groups of form $\mathbb{F} \times \mathbb{F}$

- the black-box function B is constant on the subsets

$$H_{Q,z} := \{(x, Q(x) + z) : x \in \mathbb{F}\}$$

of $\mathbb{F}^2$ and distinct for different values of z

- assume that Q is linear, i.e., $Q(X) = qX$ with unknown q
 $\Rightarrow$

$$H_Q := \{(x, Q(x)) : x \in \mathbb{F}\}$$

is a subgroup as $(x, Q(x)) + (x', Q(x')) = (x + x', Q(x, x'))$

- the sets $H_{Q,z}$ correspond to the cosets of the hidden subgroup H_Q

Classical Query Complexity

- we need to find at least n different points

$$(x_1, y_1), \dots, (x_n, y_n) \text{ with } B(x_1, y_1) = \dots B(x_n, y_n)$$

in order to determine the hidden polynomial Q of degree n

- the probability of obtaining such an n -fold collision is $1/p^{n-1}$
- $\Rightarrow$ classical query complexity is polynomial in p

Reduction to Quantum State Identification

- evaluate the black-box function on an equally weighted superposition of $(x, y) \in \mathbb{F}^2$

$$\frac{1}{p} \sum_{x, y \in \mathbb{F}} |x\rangle \otimes |y\rangle \otimes |B(x, y)\rangle$$

- measure third register; assume we have obtained $\pi(z)$, then the state on the first and second register is

$$|\phi_{Q,z}\rangle := \frac{1}{\sqrt{p}} \sum_{x \in \mathbb{F}} |x\rangle \otimes |Q(x) + z\rangle,$$

where $z \in \mathbb{F}$ is uniformly at random; the density matrix $\rho_Q := \frac{1}{p} \sum_{z \in \mathbb{F}} |\phi_{Q,z}\rangle \langle \phi_{Q,z}|$ is called a polynomial function state

Quantum Query Complexity

- let ρ and σ be two quantum state with spectral decompositions $\rho = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$ and $\sigma = \sum_j \mu_j |\phi_j\rangle\langle\phi_j|$
- assume that $\max_{i,j} |\langle\psi_i|\phi_j\rangle| \leq \alpha$ for some $\alpha \Rightarrow$

$$F(\rho, \sigma) \leq \alpha \cdot \min \left\{ \sum_i \sqrt{\lambda_i}, \sum_j \sqrt{\mu_j} \right\},$$

where $F(\sigma, \rho) := \|\sqrt{\rho}\sqrt{\sigma}\|_1$ is the fidelity of ρ and σ

- we have $F(\rho_Q, \rho_{\tilde{Q}}) \leq n/\sqrt{p}$ for any pair of different polynomial function states as

$$|\langle\phi_{Q,z}|\phi_{\tilde{Q},\tilde{z}}\rangle| = \frac{1}{p} |\{x \in \mathbb{F} : Q(x) + z = \tilde{Q}(x) + \tilde{z}\}| \leq \frac{n}{p}$$

Quantum Query Complexity

- there is a POVM $\{E_Q\}$ acting on k copies of a polynomial function state such that

$$P_{\text{success}} := \min_Q \text{Tr}(\rho_Q^{\otimes k} E_Q) \geq 1 - \epsilon$$

provided that $k \geq 2(\log N - \log \epsilon)/(-\log F)$, where

- $N := p^n$ is the number of different ρ_Q

- $F := \max_{Q \neq \tilde{Q}} F(\rho_Q, \rho_{\tilde{Q}})$

(this follows from results by HARROW AND WINTER)

- this bound and the upper bound on the maximal fidelity $F \leq n/\sqrt{p}$ imply P_{success} is at least $1/2$ for $k = 4n$
- n copies are required to have P_{success} at least $1/2$ (follows from a lower bound)

Structure of ρ_Q

- the state ρ_Q can be written as

$$\begin{aligned}\rho_Q &= \frac{1}{p^2} \sum_{b,c \in \mathbb{F}} |b\rangle\langle c| \otimes \sum_{z \in \mathbb{F}} |Q(b) + z\rangle\langle Q(c) + z| \\ &= \frac{1}{p^2} \sum_{b,c \in \mathbb{F}} |b\rangle\langle c| \otimes S^{Q(b)-Q(c)},\end{aligned}$$

where S is the cyclic shift operator acting by
 $S|u\rangle = |u + 1 \bmod p\rangle$

- apply the Fourier transform F on the second register to diagonalize all powers of S

$$\hat{\rho}_Q := (\mathbf{1} \otimes F) \rho (\mathbf{1} \otimes F)^\dagger = \frac{1}{p^2} \sum_{b,c,x \in \mathbb{F}} \omega_p^{[Q(b)-Q(c)] \cdot x} |b\rangle\langle c| \otimes |x\rangle\langle x|$$

Structure of $\hat{\rho}_Q^{\otimes k}$

$$\begin{aligned}\hat{\rho}_Q^{\otimes k} &= \frac{1}{p^{2k}} \sum_{b,c,x \in \mathbb{F}^k} \omega_p^{\sum_{j=1}^k [Q(b_j) - Q(c_j)]x_j} |b\rangle\langle c| \otimes |x\rangle\langle x| \\ &= \frac{1}{p^{2k}} \sum_{b,c,x \in \mathbb{F}^k} \omega_p^{\langle q | \Phi_n(b) - \Phi_n(c) | x \rangle} |b\rangle\langle c| \otimes |x\rangle\langle x|\end{aligned}$$

• $\langle q | := (q_1, q_2, \dots, q_n) \in \mathbb{F}^{1 \times n}$ where $Q(X) = \sum_{i=1}^n q_i X^i$



$$\Phi_n(b) := \sum_{i=1}^n \sum_{j=1}^k b_j^i |i\rangle\langle j| = \begin{pmatrix} b_1 & b_2 & \dots & b_k \\ b_1^2 & b_2^2 & \dots & b_k^2 \\ \vdots & \vdots & & \vdots \\ b_1^n & b_2^n & \dots & b_k^n \end{pmatrix},$$

• $|x\rangle := (x_1, \dots, x_k)^T \in \mathbb{F}^k$

Algebraic-Geometric Problem

- determine all $b \in \mathbb{F}^k$ for given $x \in \mathbb{F}^k$ and $w \in \mathbb{F}^n$ s.t.

$$\begin{pmatrix} b_1 & b_2 & \cdots & b_k \\ b_1^2 & b_2^2 & \cdots & b_k^2 \\ \vdots & \vdots & & \vdots \\ b_1^n & b_2^n & \cdots & b_k^n \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}$$

- denote the set of solutions and its cardinality by

$$S_w^x := \{b \in \mathbb{F}^k : \Phi_n(b)|x\rangle = |w\rangle\} \quad \text{and} \quad \eta_w^x := |S_w^x|$$

- define the quantum states $|S_w^x\rangle$ for $\eta_w^x > 0$

$$|S_w^x\rangle := \frac{1}{\sqrt{\eta_w^x}} \sum_{b \in S_w^x} |b\rangle$$

Structure of $\hat{\rho}_Q^{\otimes k}$

$$\begin{aligned}
 \hat{\rho}_Q^{\otimes k} &= \frac{1}{p^{2k}} \sum_{b, c, x \in \mathbb{F}^k} \omega_p^{\langle q | \Phi_n(b) | x \rangle - \langle q | \Phi_n(c) | x \rangle} |b\rangle \langle c| \otimes |x\rangle \langle x| \\
 &= \frac{1}{p^{2k}} \sum_{x \in \mathbb{F}^k} \sum_{w, v \in \mathbb{F}^n} \omega_p^{\langle q | w \rangle - \langle q | v \rangle} \sqrt{\eta_w^x \eta_v^x} |S_w^x\rangle \langle S_v^x| \otimes |x\rangle \langle x|
 \end{aligned}$$

Measurement of x

- the block structure implies that we can measure $x \in \mathbb{F}^k$ without loss of generality
- the probability of obtaining a particular x is $1/p^k$ (uniformly at random)
- the reduced state is

$$\hat{\rho}_Q^x := \frac{1}{p^k} \sum_{w,v \in \mathbb{F}^n} \omega_p^{\langle q|w \rangle - \langle q|v \rangle} \sqrt{\eta_w^x \eta_v^x} |S_w^x\rangle \langle S_v^x|$$

Ideal Quantum Sampling

- assume we could implement a unitary U_x with

$$U_x |S_w^x\rangle = |w\rangle$$

- applying U_x to $\hat{\rho}_Q^x$ gives

$$U_x \hat{\rho}_Q^x U_x^\dagger = \frac{1}{p^k} \sum_{w,v \in \mathbb{F}^n} \omega_p^{\langle q|w\rangle - \langle q|v\rangle} \sqrt{\eta_w^x \eta_v^x} |w\rangle \langle v|$$

Measurement in Fourier basis

- measure in the Fourier basis, i.e., carry out the orthogonal measurement with respect to the states

$$|\psi_Q\rangle := \frac{1}{\sqrt{p^n}} \sum_{w \in \mathbb{F}^n} \omega_p^{\langle q|w\rangle} |w\rangle$$

- the probability for correct detection of $\hat{\rho}_Q^x$ is

$$\langle \psi_Q | U_x \hat{\rho}_Q^x U_x^\dagger | \psi_Q \rangle = \frac{1}{p^{k+n}} \left(\sum_{w \in \mathbb{F}^n} \sqrt{\eta_w^x} \right)^2$$

Summery of the Steps

- apply Fourier on second register
- measure x in second register
- apply (ideal) quantum sampling U_x
- measure in Fourier basis
- the outcome determines the coefficients of the hidden polynomial
- the overall success probability is

$$\frac{1}{p^{2k+n}} \sum_{x \in \mathbb{F}^k} \left(\sum_{w \in \mathbb{F}_p^n} \sqrt{\eta_w^x} \right)^2$$

Approximate Quantum Sampling

- determine all $b \in \mathbb{F}^k$ for given $x \in \mathbb{F}^k$ and $w \in \mathbb{F}^n$

$$\begin{pmatrix} b_1 & b_2 & \cdots & b_k \\ b_1^2 & b_2^2 & \cdots & b_k^2 \\ \vdots & \vdots & & \vdots \\ b_1^n & b_2^n & \cdots & b_k^n \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}$$

- recall $S_w^x := \{b \in \mathbb{F}^k : \Phi_n(b)|x\rangle = |w\rangle\}$ and $\eta_w^x := |S_w^x|$
- it suffices to implement an approximate version V_x of quantum sampling satisfying

$$V_x|S_w^x\rangle = |w\rangle$$

for a vast majority of pairs (x, w) .

Properties of Generic Morphisms

Proposition

- consider a morphism $f : \mathbb{A}^k \times \mathbb{A}^n \rightarrow \mathbb{A}^n$ defined over $\mathbb{Z}$
 - f is given by an n -tuple $f = (f_1, \dots, f_n)$ of polynomials in $\mathbb{Z}[X, B]$
 - $X = (X_1, \dots, X_k)$ and $B = (B_1, \dots, B_n)$ are the coordinates on $\mathbb{A}^k$ and on the first copy of $\mathbb{A}^n$
- suppose that the Jacobian determinant $\det(\partial f_i / \partial B_j)_{ij}$ is a non-zero element of $\mathbb{Z}[X, B]$
- $\Rightarrow$ there exist a positive real number γ and a non-zero polynomial $g \in \mathbb{Z}[X]$ such that for all finite fields $\mathbb{F}$ and $x \in \mathbb{F}^k$ with $g(x) \neq 0$ we have $|f(\{x\} \times \mathbb{F}^n)| \geq \gamma |\mathbb{F}|^n$

Implications for our AG problem

- let $X_{\text{good}} := \{x \in \mathbb{F}^k : g(x) \neq 0\}$
- for all $x \in X_{\text{good}}$ let $W_{\text{good}}^x \subseteq \mathbb{F}^n$ to the set such that

$$w \in W_{\text{good}}^x \Rightarrow 1 \leq \eta_x^w \leq d$$

where d is some constant independent of the size of $\mathbb{F}$

- we can implement quantum sampling

$$V_x |S_w^x\rangle = |w\rangle$$

for all (x, w) with $x \in X_{\text{good}}$ and $w \in W_{\text{good}}^x$ because

- S_w^x have constant size for all good pairs (x, w) and
- all solutions $b \in S_w^x$ can be enumerated efficiently

Success of Approximate Measurement

if choose the number of copies k to be equal to n (the maximal degree of the hidden polynomial) then

- $|X_{\text{good}}| = \Omega(p^n)$
- $|W_{\text{good}}^x| = \Omega(p^n)$
- the overall success probability is bounded from below as follows

$$\begin{aligned} P_{\text{success}} &= \frac{1}{p^{3n}} \sum_{x \in X_{\text{good}}} \left(\sum_{w \in W_{\text{good}}^x} \sqrt{\eta_w^x} \right)^2 \\ &\approx \frac{1}{p^{3n}} \Omega(p^n) \left(\Omega(p^n) \right)^2 = \Omega(1) \end{aligned}$$

Conclusions and Possible Extensions

- definition of a new black-box problem (a special case of the hidden polynomial problem)
- classically hard (polynomial classical query complexity)
- polylogarithmic quantum algorithm (for fixed degree)
- $\Rightarrow$ what useful real-life problem can be reduced to this black-box problem???
- possible extensions:
 - polynomials over non-prime fields; this is straightforward
 - polynomials over rings such as $\mathbb{Z}_d$???
 - multivariate polynomials??? some cases can be easily reduced to univariate case, e.g., hidden hyperbola